CR510 - Teoremi per l'esame

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1 Struttura di gruppo sui punti definiti su un campo K di una cubica piana liscia e proiettica con un punto K-razionale (Teorema 1.2.1)

Theorem 1.1. (Teorema di Bezout) Let C_1 and C_2 be two projective curves over a field K of degree d and e respectively. Then

$$\sum_{P \in C_1(\overline{K}) \cap C_2(\overline{K})} I(P, C_1 \cap C_2) = de.$$

In other words C_1 and C_2 intersect over $\overline{K}$ in exactly de points, counting multiplicities.

As an application we can use Bezout Theorem to compute the intersection of the projective curve $\mathcal{C}$ given by

$$y^2z - x^3 - Axz^2 - Bz^3$$

with the line at infinity $L_\infty : z = 0$. We have already seen that the intersection is a single point $P = (0 : 1 : 0)$. Since the curve has degree 3 and the line has degree 1, Bezout's Theorem implies that the intersection has multiplicity 3. This can be computed explicitly as follows: we can work in the chart $y \neq 0$ in which the point P becomes the origin. Then,

$$I(P, L_\infty \cap \mathcal{C}) = I(z, z - x^3 - Axz^2 - Bz^3) = I(z, x^3) = 3.$$

By the theorem of Bezout, we can define the following operation:

$$\oplus : E(K) \times E(K) \to E(K)(P, Q) \mapsto P \oplus Q$$

where $P \oplus Q$ is the third point of intersection of $\ell_{PQ} \cap C$ where R is the third point of intersection between $l_{PQ} \cap \mathcal{C}$.

Lemma 1.2. Given A, B two non-singular and irreducible curves such that $\#(A \cap B) = 8$ and a third non singular and irreducible curve C such that goes trough at least 8 of the 9 point of intersection between $A \cap B$; then, C goes trough the 9th point of intersection.

Theorem 1.3. Given a non singular projective curve C of degree three defined over a field K, and a point $\mathcal{O} \in C(K)$, the operation $\oplus$ defines a group $(C(K), \oplus)$ such that:

1. The group is commutative, i.e. $P \oplus Q = Q \oplus P$ for every $P, Q \in C(K)$;
2. The point $\mathcal{O}$ is the identity element, i.e. $P \oplus \mathcal{O} = P$ for every $P \in C(K)$;
3. Every point has its inverse, i.e. for every $P \in C(K)$ there exists a point $-P \in C(K)$ such that $P \oplus (-P) = \mathcal{O}$;
4. The operation $\oplus$ is associative, i.e. for evey $P, Q, R \in \mathcal{C}(K)$ we have $(P \oplus Q) \oplus R = P \oplus (Q \oplus R)$

Proof. From the definition of $\oplus$ we can see that it is commutative, since the line ℓ_{PQ} and ℓ_{QP} coincide. This proves (1).

Let us compute $P \oplus \mathcal{O}$: by definition we define $R := P * Q$ to be the third point of intersection $\ell_{PQ} \cap \mathcal{C}$, and then $P \oplus Q$ is $R * \mathcal{O}$, i.e. the third point of intersection $\ell_{(P*Q)\mathcal{O}} \cap \mathcal{C}$. But by construction P, $P * \mathcal{O}$ and $\mathcal{O}$ are collinear, i.e. $\ell_{P\mathcal{O}} = \ell_{(P*\mathcal{O})\mathcal{O}}$. Therefore $P \oplus \mathcal{O} = P$ as wanted (note that the argument did not require $P * Q$ to be distinct from P or $\mathcal{O}$). This proves (2).

To prove that every point P has an inverse we will show that $-P$ is $P * (\mathcal{O} * \mathcal{O})$, i.e. is the third point of intersection $\ell_{P(\mathcal{O}*\mathcal{O})} \cap \mathcal{C}$, where $\mathcal{O} * \mathcal{O}$ is the third point of intersection of the tangent line at $\mathcal{C}$ in $\mathcal{O}$ (we do not exclude that $\mathcal{O} * \mathcal{O} = \mathcal{O}$). Now, to compute $P \oplus (-P)$ we need the line $\ell_{P(-P)}$ that by construction coincide with the line $\ell_{P(\mathcal{O}*\mathcal{O})}$, since the points $P, \mathcal{O} * \mathcal{O}, (-P)$ are collinear. But then $P \oplus (-P)$ will be the third point of intersection $\ell_{(\mathcal{O}*\mathcal{O})(-P)} \cap \mathcal{C}$, and, by construction, the line $\ell_{(\mathcal{O}*\mathcal{O})\mathcal{O}}$ is the tangent line at $\mathcal{C}$ in $\mathcal{O}$ so that $P \oplus (-P) = \mathcal{O}$ as wanted. This proves (3).

Property (4) is the hardest to show. There are three main ways to prove associativity: the first one is by hand by applying repetitively the definition of the operation. The second one uses the Riemann–Roch Theorem, and gets the result as an immediate corollary of a different characterization of the group operation. Finally, the third way, which we will present here, uses a more geometric approach.

The goal is to show that $(P \oplus Q) \oplus R = P \oplus (Q \oplus R)$. It is then sufficient to show that $(P \oplus Q) * R = P * (Q \oplus R)$ i.e. the third point of intersection of $\ell_{(P\oplus Q)R} \cap \mathcal{C}$ and $\ell_{P(Q\oplus R)} \cap \mathcal{C}$ coincide. Let us define the following six lines: $\square$

2 Isomorfismo tra i gruppo E(K) e Pic0(E) e nuova dimostrazione della struttura di gruppo (Proposizione 1.2.26)

Definition 2.1. $\mathrm{Pic}^0(\mathcal{C}) = \frac{\mathrm{Div}^0(\mathcal{C})}{\mathrm{Prin}(C)}$

Definition 2.2. Given a divisor $D \in \mathrm{Div}(\mathcal{C})$ we define

$$L(D) := \{\varphi \in K(\mathcal{C}) \,:\, \mathrm{div}\varphi + D \geq 0\} \cup \{0\}$$

the vector space. We also define $l(D) := \dim_K L(D)$ the dimension as $K-$vector space.

Lemma 2.3. Let $\mathcal{C}$ be a non singular projective curve of genus 1. Then for every divisor D with $\deg D \geq 1$ one has

$$l(D) = deg(D)$$

Lemma 2.4. Let $\mathcal{C}$ a non singular projective curve of genus 1, and let $f \in K(\mathcal{C})$. If $\mathrm{div} f = [P] - [Q]$ then f is constant, end hence $P = Q$.

Theorem 2.5. Let $\mathcal{C}$ be a non singular projective curve of genus 1 and let $\mathcal{O} \in \mathcal{C}(K)$. Then the map

$$\begin{array}{rcl} \mathcal{C}(K): & \longrightarrow & \mathrm{Pic}^0(\mathcal{C}) \\ P & \longmapsto & \overline{[P] - [\mathcal{O}]} \end{array}$$

Proof. First we show that the map is injective. Assume that there are two points $P, Q \in C(K)$ such that

$$\overline{[P] - [\mathcal{O}]} = \overline{[Q] - [\mathcal{O}]} \text{ in } \mathrm{Pic}^0(C)$$

this implies that

$$[P] - [\mathcal{O}] \sim [Q] - [\mathcal{O}],$$

i.e., there exists a rational function $f \in K(C)$ such that $\mathrm{div}\, f = [P] - [\mathcal{O}] - [Q] + [\mathcal{O}] = [P] - [Q]$. By lemma 2.3 we conclude that $[P] = [Q]$ as wanted.
Let's show that the map is injective. Let us fix a (class of a) degree zero divisor $\overline{D} \in \mathrm{Pic}^0(C)$ (then $\deg(D) = 0\ \forall D \in \overline{D}$). By the Riemann–Roch Theorem for genus 1 curves, lemma 2.4, since $D + [\mathcal{O}] \geq 0$ and $\deg(D + [\mathcal{O}]) = 1$ we have that $\ell(D + [\mathcal{O}]) = 1$. We choose a nonzero element $f \in L(D + [\mathcal{O}])$ (then $\mathrm{div} f + D + [\mathcal{O}] \geq 0$), which is therefore a basis for the vector space. Note that, since $f \in K(C)$, $\deg \mathrm{div}\, f = 0$, and f satisfies $\mathrm{div}\, f + D + [\mathcal{O}] \geq 0$. In particular $\mathrm{div}\, f + D + [\mathcal{O}]$ is a positive divisor of degree 1, so is of the form $[P]$ for some $P \in C(K)$. But then

$$[P] = \mathrm{div}\, f + D + [\mathcal{O}] \Rightarrow D \sim [P] - [\mathcal{O}],$$

so the map is surjective. □

We can use the bijection of theorem 2.5 to define a group structure on $C(K)$ as follows: given $P, Q, R \in C(K)$ we have $P + Q = R$ if and only if

$$[P] - [\mathcal{O}] + [Q] - [\mathcal{O}] = [R] - [\mathcal{O}] \iff [P] + [Q] = [R] + [\mathcal{O}]$$

using the operation in $\mathrm{Pic}^0(C)$. In fact we can show that this operation is the same we have defined using the chord and tangent construction. To see this let $P, Q, R \in C(K)$ such that $P \oplus Q = R$. Recall that this means that $R = (P * Q) * \mathcal{O}$, i.e., R is the third point of intersection of $\ell_{P*Q,\mathcal{O}} \cap C$, where $P * Q$ is the third point of intersection of $\ell_{PQ} \cap C$. Denote $\ell_1 = \ell_{PQ}$ and $\ell_2 = \ell_{(P*Q)\mathcal{O}}$. Then we can define

$$\varphi = \ell_1/\ell_2 \in K(C),$$

where we identify the lines with their equations. Since we know the intersections $\ell_1 \cap C$ and $\ell_2 \cap C$ we can compute the divisor associated to φ as

$$\operatorname{div} \varphi = [P] + [Q] + [P * Q] - [P * Q] - [\mathcal{O}] - [R] = [P] + [Q] - ([R] + [\mathcal{O}]).$$

In particular

$$[P] + [Q] \sim [R] + [\mathcal{O}]$$

so that $P + Q = R$ using the operation induced by $\mathrm{Pic}^0(C)$.
Note that, using theorem 2.5, we do not need to verify that the operation satisfies all the axiom of the group, since *by definition*, $\mathrm{Pic}^0(C)$ is a group! On the other hand the association we have defined is bijective, hence the operation we define in this way is also a group operation.

3 Esistenza dell'equazione di Weierstrass per curve lisce proiettive di genere 1 (Teorema 1.2.27)

Lemma 3.1. LEt $\phi : \mathcal{C}_1 \to \mathcal{C}_2$ be a non constant map of non singular projective curves. Assume that the field extension $K(\mathcal{C}_1) \supset \phi^*(K(\mathcal{C}_2))$ is separable. Then for all but finitely many $Q \in \mathcal{C}_2(K)$

$$\#\phi^{-1}(Q) = deg\phi$$

Morevoer, if deg$\phi = 1$ then ϕ is an isomorphism.

Theorem 3.2. Let E be a non singular projective curve of genus 1 over a field K with a point $\mathcal{O} \in E(K)$ then there is an isomorphism between E and $\mathcal{C}$ where $\mathcal{C}$ is defined by

$$y^2 + a_1xy + a_3y^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3.$$

Proof. Since E has genus 1, lemma 2.3 shows that $\ell(m\mathcal{O}) = m$ for every $m \geq 1$. Since the constant functions lie in $L(\mathcal{O})$ and $l([O]) = \dim L([\mathcal{O}]) = 1$ we have that $L([\mathcal{O}]) = K$, and $\{1\}$ is a basis. Since $\ell([2\mathcal{O}]) = 2$ there exists a non constant function $x \in K(E)$ such that $\{1, x\}$ is a basis of $L(2[\mathcal{O}])$ and div$x \notin L[\mathcal{O}]$. Similarly we can choose a function $y \in L(3[\mathcal{O}])$ so that $\{1, x, y\}$ is a basis of $L(3[\mathcal{O}])$. Then, $\{1, x, x^2, y\}$ is a basis of $L(4[\mathcal{O}])$ [1], and $\{1, x, x^2, y, xy\}$ is a basis of $L(5[\mathcal{O}])$.
The set $\{1, x, x^2, y, xy, x^3, y^2\}$ is contained in $L(6[\mathcal{O}])$, which is a K-vector space of dimension 6. This implies that the seven functions have to be linearly dependent, i.e. there exists $a_i \in K$ such that

$$a_0y^2 + a_1xy + a_3y = a_0'x^3 + a_2x^2 + a_4x + a_6,$$

and the equality holds as functions, regular outside of the only pole $\mathcal{O}$. Note that, since $\{1, x, x^2, y, xy\}$ are linearly independent, a_0 and a_0' are both different from zero (at least one has to be for dimension reason, and if exactly one was zero, the other will give a linear combination of functions in $L(5[\mathcal{O}])$ equal a function that has a pole of order strictly larger than 5 at $\mathcal{O}$). Therefore, after replacing y with a_0y/a_0' and x with a_0x/a_0', and multiplying all the terms by $a_0'^2/a_0^3$ we can suppose $a_0 = a_0' = 1$. We define the map

$$E \setminus \{\mathcal{O}\} \to \mathcal{C} \quad \text{given by } P \mapsto (x(P), y(P))$$

which maps the affine curve $E \setminus \{\mathcal{O}\}$ into the plane affine curve $\mathcal{C}$ with equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

We now show that the map $P \mapsto (x(P), y(P))$ is an isomorphism. First we consider the rational function $x \in K(E)$: by construction it has a single pole of order exactly two at $\mathcal{O}$, and therefore only two zeros. We claim that the composition

$$E \setminus \{\mathcal{O}\} \to \mathcal{C} \to \mathbb{A}^1 \quad P \mapsto (x(P), y(P)) \mapsto x(P)$$

[1] We have that $divx^2 = divx + divx \geq -4[\mathcal{O}]$, but also that $divx^2 < -3[\mathcal{O}]$ since $divx \notin L[\mathcal{O}]$. Then $[x^2] \in L(4[\mathcal{O}]) \setminus L(3[\mathcal{O}])$.

is $2:1$. So in particular, by lemma 3.1, $degx = 2$. This follows from the fact that we can compute the cardinality of the fiber of any point in $\mathbb{A}^1$, and we have that $|x^{-1}(0)| = |\{P \in E \,|\, x(P) = 0\}| = 2$, just as we said before. Similarly the map defined using the rational function $y \in K(E)$ has $degy = 3$ since it is almost always $3:1$. To conclude is enough to observe that the degree of the map $P \mapsto (x(P), y(P))$ has to divide both $degx$ and $degy$[2] which implies that it is a map of degree 1 and hence, by lemma 3.1, an isomorphism. $\square$

[2]This comes from 3.1, indeed we have that $deg(g \circ f) = \sum_{y \in g^{-1}(z)} \sum_{x \in f^{-1}(y)} 1 = \deg g \deg f$.

4 Due curve ellittiche sono isomorfe se e solo se hanno lo stesso j-invariante; dato $j_0 \in K$ esiste sempre una curva ellittica definita su K con j-invariante j_0 (Teorema 1.2.32)

Lemma 4.1. Let $E_{A,B}$ and $E_{A',B'}$ be two elliptic curves in short Weierstrass form, i.e. $y^2 = x^3 * Ax + B$. If

$$\varphi : E_{A',B'} \to E_{A,B}, \qquad \varphi(\mathcal{O}) = \mathcal{O},$$

is an isomorphism, then there exists $c \in K^\times$ such that $A' = c^4 A$, $B' = c^6 B$ and the map φ is given by

$$(x : y : z) \mapsto (c^2 x : c^3 y : z).$$

Conversely, if $A' = c^4 A$ and $B' = c^6 B$ for some $c \in K^\times$, then the map $\varphi(x : y : z) = (c^2 x : c^3 y : z)$ is an isomorphism between the curve $E_{A',B'}$ and $E_{A,B}$ that sends $\mathcal{O}$ to $\mathcal{O}$.

Definition 4.2. Given an elliptic curve over a field K, with char$K \neq 2, 3$ with Weierstrass equation $E_{A,B}$ the **j-invariant** of the elliptic curve is defined as

$$j(E) := \frac{1728(4A^3)}{4A^3 + 27B^2}$$

where $\Delta := 4A^3 + 27B^2$ is the discriminant of $E_{A,B}$.

Theorem 4.3. Given two elliptic curves E_1, E_2 over a field K of char $K \neq 2, 3$, then E_1 is isomorphic to E_2 over K if and only if $j(E_1) = j(E_2)$; moreover, given $j_0 \in \overline{K}$, there exists E_0 elliptic curve over $\overline{K}$ such that $j(E_0) = j_0$.

Proof. ($\Rightarrow$)If E_1 and E_2 are isomorphic, then we can apply lemma 4.1 and conclude that the two j-invariants are equal. Indeed, we have $A' = c^4 A$ and $B' c^6 B$ and

$$j(E') = \frac{1728(4A'^3)}{4A'^3 + 27B'^2} = \frac{1728(4c^{12}A^3)}{4c^{12}A^3 + 27c^{12}B^2} = \frac{1728(4A^3)}{4A^3 + 27B^2} = J(E)$$

($\Leftarrow$)Conversely suppose that the two j-invariants are equal. Let us fix Weierstrass equation for the two curves:

$$E_1 : y^2 = x^3 + A_1 x + B_1 \qquad E_2 : y^2 = x^3 + A_2 x + B_2.$$

The hypothesis $j(E_1) = j(E_2)$ yields

$$4(A_1 A_2)^3 + 27A_1^3 B_2^2 = 4(A_1 A_2)^3 + 27A_2^3 B_1^2,$$

so in particular we get $A_1^3 B_2^2 = A_2^3 B_1^2$. Together with the fact that the discriminant of both equations is different from zero (they are both elliptic curves), it follows that $A_1 = 0$ if and only if $A_2 = 0$ and $B_1 B_2 \neq 0$, and similarly $B_1 = 0$ if and only if $B_2 = 0$ and $A_1 A_2 \neq 0$.
By lemma 4.1, we know that to give an isomorphism between E_1 and E_2 it is sufficient to determine an element $c \in K^\times$ such that $A_2 = c^4 A_1$ and $B_2 = c^6 B_1$. We distinguish three cases:

$A_1 = 0$

As noted above this implies $A_2 = 0$ and $B_1 B_2 \neq 0$ (note that the only value of j in this case is $j = 0$). If we had an isomorphism $E_1 \to E_2$ this will send $x \mapsto c^2 x$ and $y \mapsto c^3 y$. This maps the curve E_1 into the curve of equation

$$y^2 = x^3 + \frac{B_1}{c^6}.$$

If we choose c such that $c^6 = B_1/B_2$ then we obtain the equation of E_2 as wanted.

$B_1 = 0$

The same argument as above shows that we can choose c such that $c^4 = A_1/A_2$.

$A_1 B_1 \neq 0$

In this case the substitution yields

$$y^2 = x^3 + \frac{A_1}{c^4} x + \frac{B_1}{c^6}.$$

In this case we want $c^4 = A_2/A_1$ and $c^6 = B_2/B_1$. Such a $c \in K^\times$ exists since the condition $j(E_1) = j(E_2)$ implies

$$\left(\frac{A_2}{A_1}\right)^3 = \left(\frac{B_2}{B_1}\right)^2.$$

$\square$

5 Forma standard di un'isogenia e descrizione della cardinalità del nucleo a partire dalla forma standard (Lemma 2.3.1-2.3.3 e Corollario 2.3.6)

Lemma 5.1. Let $\alpha = \left(\frac{u}{v}, \frac{s}{t}y\right)$ an isogeny between E_1 and E_2, respectively defined by $y^2 = f_1(x)$ and $y^2 = f_2(x)$. Then $v^3|t^2|(v^3 f_1)$; moreover v and t have the same set of roots in $\overline{K}$.

Lemma 5.2. *Let E be an elliptic curve over a field K in short Weierstrass form*

$$E : y^2 = x^3 + Ax + B$$

and let $f \in K(E)$ be a rational function on E. We say that f is even if $f(P) = f(-P)$ for every $P \in E(K)$. Then f is even if and only if $f \in K(x)$.[3]

Proof. We know by the property of the group operation that if $P = (x_0, y_0)$ then $-P = (x_0, -y_0)$ so that every element of $\overline{K}(x)$ is even. Let us now suppose that $f \in K(E)$ is even. Using the Weierstrass equation we can write f as

$$f(x, y) = g(x) + h(x)y \qquad \text{with } g, h \in \overline{K}(x).$$

Then the condition $f(P) = f(-P)$ implies that

$$\begin{aligned} f(x, y) &= f(x, -y) \\ g(x) + h(x)y &= g(x) - h(x)y \\ 2yh(x) &= 0 \end{aligned}$$

for every x, y. Hence if char $K \neq 2$ this holds only if h is the zero polynomial. □

Lemma 5.3. LetE_1 and E_2 be elliptic curves defined over K in short Weierstrass form and let $\alpha : E_1 \to E_2$ an isogeny. Then, α can be defined by an affine rational map of the form

$$\alpha(x, y) = \left(\frac{u(x)}{v(x)}, \frac{s(x)}{t(x)}y\right)$$

with $u, v, s, t \in K[x]$ and $gcd(u, v) = gcd(s, t) = 1$.

Proof. Assume that α is defined by the rational map $[\alpha_x : \alpha_y : \alpha_z]$; then, for any affine point $[x : y : 1] \in E_1(\overline{K})$ we can write

$$\alpha(x, y) = (r_1(x, y), r_2(x, y)),$$

where

$$r_1(x, y) = \frac{\alpha_x(x, y, 1)}{\alpha_z(x, y, 1)}, \quad r_2(x, y) = \frac{\alpha_y(x, y, 1)}{\alpha_z(x, y, 1)}.$$

[3] It is easy to see that all the arguments of this section apply readily to curves of the form $y^2 = f(x)$, so we need only to assume that char$(K) \neq 2$.

By repeatedly using the equation $y^2 = x^3 + Ax + B$ to replace y^2 with a polynomial in x, we can assume that both r_1 and r_2 have degree at most 1 in y. We then have

$$r_1(x, y) = \frac{p_1(x) + p_2(x)y}{p_3(x) + p_4(x)y}$$

for some $p_1, p_2, p_3, p_4 \in K[x]$. We now multiply the numerator and denominator of $r_1(x, y)$ by $p_3(x) - p_4(x)y$ and use the curve equation to get rid of y^2 in the denominator, and we have

$$r_1(x, y) = \frac{q_1(x) + q_2(x)y}{q_3(x)}$$

for some $q_1, q_2, q_3 \in K[x]$. We now use the property that α is a group homomorphism, hence $\alpha(-P) = -\alpha(P)$ for any $P \in E(\overline{K})$. Recall that the inverse of an affine point $P = (x, y)$ on a curve in short Weierstrass form is $(x, -y)$; thus, if $-\alpha(x, y) = \alpha(x, -y)$, we have

$$\begin{aligned}(r_1(x, y), -r_2(x, y)) &= -(r_1(x, y), r_2(x, y)) = -\alpha(x, y) \\ &= \alpha(-(x, y)) = (r_1(-(x, y)), r_2((-x, y)))\end{aligned}$$

Consequently, $r_1(x, y)$ is an even rational function in $K(E)$. By Lemma 2.3.3 we conclude that $q_2 = 0$. After eliminating any common factor between q_1 and q_3, we set $\frac{q_1(x)}{q_3(x)} = \frac{u(x)}{v(x)}$ with $(u, v) = 1$. For $r_2(x, y)$, we now use $r_2(x, y) = -r_2(x, -y)$, indeed if we have $r_2(x, y) = \frac{q_1(x)+q_2(x)y}{q_3(x)}$, then we have:

$$-\frac{q_1(x) + q_2(x)y}{q_3(x)} = \frac{q_1(x) - q_2(x)y}{q_3(x)}$$

hence

$$2\frac{q_1(x)}{q_3(x)} = 0 \implies q_1(x) = 0$$

□

Corollary 5.4. *Let $\alpha(x, y) = \left(\frac{u(x)}{v(x)}, \frac{s(x)}{t(x)}y\right)$ be an isogeny $E_1 \to E_2$ in standard form; then, the affine points $[x_0 : y_0 : 1] \in E_1(\overline{K})$ in the kernel of α are exactly those for which $v(x_0) = 0$.*

Proof. ($\Rightarrow$) If $v(x_0) \neq 0$, then $t(x_0) \neq 0$ and $\alpha(x_0, y_0) = \left(\frac{u(x_0)}{v(x_0)}, \frac{s(x_0)}{t(x_0)}y_0\right)$ is an affine point and therefore not $\mathcal{O}$, hence $(x_0, y_0) \notin \ker(\alpha)$.
($\Leftarrow$) By homogenizing and putting α into projective form, we can write α as

$$\alpha = [ut : vsy : vt],$$

where ut, vs and vl are now homogeneous polynomials of equal degree ($s, t, u, v \in K[x, z]$). Suppose $y_0 \neq 0$; by the previous lemma, if $v(x_0, 1) = 0$, then $t(x_0, 1) = 0$ and since $v^3 \mid t^2$ (by lemma 5.1), the multiplicity of $(x_0, 1)$ as a root of l is strictly greater than its multiplicity as a root of v [4] . This implies that, working over K, we can re-normalize α by dividing by

[4]Since we know that $v(x)^3 | t(x)^2$, if $(x - x_0)^m | v(x)$ (with m the algebraic multiplicity of v), we have that $(x - x_0)^{3m} | t(x)^2$. Hence, if if $(x - x_0)^n | t(x)$ (with n the algebraic multiplicity of t), we have that $2n \leq 3m \Rightarrow n > \frac{3}{2}m$. Since $n, m \in \mathbb{N}$ we have $n > m$.

a suitable power of $x - x_0$ so that α_y does not vanish at $[x_0 : y_0 : 1]$ but α_x, α_z both do [5] . Then, $\alpha(x_0 : y_0 : 1) = (0 : 1 : 0) = \mathcal{O}$, hence $(x_0 : y_0 : 1) \in \ker(\alpha)$ as claimed.
If $y_0 = 0$, then x_0 is a root of the cubic polynomial f_1 appearing in the equation of E_1 : $y^2 = f_1(x)$, and it is not a double root since E_1 is non-singular. In this case, we normalize $\alpha = [ut : vsy : vt]$ by multiplying by yz and then replacing y^2z by $f_1(x, z)$ to obtain

$$\alpha = [utyz : vsf_1 : vtyz].$$

Notice now that $(x_0, 1)$ is a root of multiplicity 1 of $f_1(x, z)$; hence, using the fact that $v^3 \mid t^2$, we have that the multiplicity of $(x_0, 1)$ as a root of vf_1 is not greater than the multiplicity as a root of t, then we have the multiplicity of $(x_0, 1)$ as root of vf_1 is smaller than the multiplicity as a root of vt; hence, we can again normalize α by dividing by a suitable power of $x - x_0z$ so that α_y does not vanish at $(x_0 : 0 : 1)$ but α_x and α_z do[6]. Thus, $[x_0 : 0 : 1]$ is again in the kernel of α, concluding the theorem. $\square$

[5] We can also suppose that $s(x_0) \neq 0$, indeed if $s(x_0) = 0$ then $\alpha(x_0, y_0)$ is an affine point and hence is not $\mathcal{O}$.

[6] The firs coordinate of α, i.e. $(utyz)$, vanishes at $(x_0 : y_0 : 1)$ since $y_0 = 0$.

6 La cardinalità del nucleo di un'isogenia è uguale al grado di separabilità dell'isogenia (Teorema 2.5.7)

Lemma 6.1. Let α be an isogeny of elliptic curves over a field K of characteristic $p > 0$; then

$$\alpha = \alpha_{\text{sep}} \circ \pi^n$$

where α_{sep} is a separable isogeny, $n \geq 0$ is an integer and π is the p-power endomorphism $\pi[x : y : z] = [x^p : y^p : z^p]$. In particouar, we have that $\deg\alpha = p^n \deg\alpha_{\text{sep}}$.

Theorem 6.2. The cardinality of the kernel of an isogeny is equal to its separable degree.

Proof. Let $\alpha = \alpha_{\text{sep}} \circ \pi^n$ then, $\#ker(\alpha) = \ker(\alpha_{\text{sep}})$ since the kernel of π (and hence of π^n) is trivial. It is thus sufficient to prove the result for separable isogenies. Assume now that α is separable and let $\left(\frac{u(x)}{v(x)}, \frac{s(x)}{t(x)}y\right)$ be its standard form. By definition

$$\ker(\alpha) = \alpha^{-1}([0 : 1 : 0]),$$

and since α is a group homomorphism, the cardinality of the preimage of a point is always the same [7] (if this set is not empty), hence it will be sufficient to compute $\#\alpha^{-1}(a, b)$ for any $(a, b) \in \alpha(E_1(K))$. Let us choose point $(a, b) \in \alpha(E_1(K))$ with $a, b \neq 0$ and such that a is not equal to the ratio of the leading coefficients of u and v. Call the set $S(a, b) := \alpha^{-1}(a, b)$. We require moreover that, for every $(x_0, y_0) \in S(a, b)$, $(u/v)'(x_0) \neq 0$, which is again possible since α is a separable isogeny, hence $(u/v)'$ is not identically zero, hence it has only finitely many roots. Let us now compute the cardinality of $S(a, b)$. If $(x_0, y_0) \in S(a, b)$, then

$$\frac{u(x_0)}{v(x_0)} = a, \qquad \frac{s(x_0)}{t(x_0)} y_0 = b.$$

We must have $t(x_0) \neq 0$, since α is defined at (x_0, y_0), and $b \neq 0$ implies $s(x_0) \neq 0$. It follows that $y_0 = \frac{t(x_0)}{s(x_0)} b$ is uniquely determined by x_0. Thus, to compute $\#S(a, b)$ it suffices to count the number of distinct values of x_0 that occur among the points in $S(a, b)$. We now let $g(x) = u(x) - av(x)$ so that $\alpha(x_0, y_0) = (a, b)$ if and only if $g(x_0) = 0$. Since a is not equal to the ratio of the leading coefficients of u and v, then

$$\deg(g) = \max\{\deg(u), \deg(v)\} = \deg(\alpha),$$

hence the cardinality of $S(a, b)$ is equal to the number of *distinct* roots of g. We are left to prove that g has all distinct roots. Any $x_0 \in K$ is a multiple root of g if and only if $g(x_0) = g'(x_0) = 0$, equivalently $av(x_0) = u(x_0)$ and $av'(x_0) = u'(x_0)$. If we multiply opposite sides of these equations and cancel the a's we get $u'(x_0)v(x_0) = u(x_0)v'(x_0)$. But by construction for every $(x_0, y_0) \in S(a, b)$ we have that $(u/v)'(x_0) \neq 0$, hence

$$u'(x_0)v(x_0) - u(x_0)v'(x_0) \neq 0,$$

[7] Let $f : G \to H$ an homomorphism group and $h \in H$ such that exists $x \in f^{-1}(h)$. Now $f^{-1}(h) = \{g \in G \mid f(g) = h\} = \{g \in G \mid f(g) = f(x)\} = \{g \in G \mid f(g - x) = 0\}$. We can define the map $\psi : f^{-1}(h) \to \ker f$ such that $g \mapsto g - x$, that is clearly bijective. Hence we have that for all h, $\#f^{-1}(h) = \#\ker f$.

which implies that g has no multiple roots, concluding the proof since

$$\#\ker(\alpha) = \#S(a,b) = \deg(g) = \deg(\alpha).$$

$\square$

7 Teorema di corrispondenza tra sottogruppi finiti dei punti di una curva ellittica e isogenie (Teorema 2.6.1)

Lemma 7.1. Let $\mathbb{K} \subset K(C_1)$ be a subfield of finite index containing K; then, there exist a smooth curve $\mathcal{C}'/K$, unique up to K-isomorphism, and a non-constant map $\phi : \mathcal{C} \to \mathcal{C}'$ defined over K such that $\phi^*(K(\mathcal{C}')) = \mathbb{K}$.

Lemma 7.2. Let $\phi : E_1 \to E_2$ an isogeny, then

$$\ker\phi \cong \mathrm{Aut}\left(\overline{K}(E_1)/\phi^*(\overline{K}(E_2))\right)$$

Lemma 7.3. (Artin's lemma) Let L a field, $G < \mathrm{Aut}(L)$ a finite subgroup and $L^G := \{f \in L \,|\, \sigma(f)\, \forall \sigma \in G\}$. Then L/L^G is a finite Galois extension and $G = \mathrm{Gal}(L/L^G)$

Lemma 7.4. Let $\phi : \mathcal{C}_1 \to \mathcal{C}_2$ a non-constant, unramified and separable morphism of curves, then

$$2(g(\mathcal{C}_1) - 1) = \deg\phi(2(g(\mathcal{C}_2) - 1))$$

Theorem 7.5. Let E/K be an elliptic curve and let G be a finite subgroup of $E(K)$; there exists an elliptic curve E_0 ans a separable isogeny $\phi : E \to E_0$ with $ker(\phi) = G$. The curve E_0 and the isogeny are defined over a finite extension of K and are unique up to isomorphism.

Proof. Let $T \in E(\overline{K})$ be a point on our elliptic curve; we can consider the *translation-by-T* map $\tau_T : E(\overline{K}) \to E(\overline{K})$ sending $P \mapsto P + T$. Let us consider the induced map of the function field of E, i.e. $\tau_T^* : \overline{K}(E) \to \overline{K}(E)$ given by $f \mapsto f \circ \tau_T$, and this is an automorphism of $\overline{K}(E)$. Let us define $\overline{G} := \{\tau_T^* \,|\, T \in G\} < \mathrm{Aut}(\overline{K}(E))$ and

$$\overline{K}(E)^{\overline{G}} := \{f \in \overline{K}(E) \mid \tau_T^*(f) = f \text{ for all } T \in G\}.$$

Since $G \cong \overline{G}$[8] and $\#G < +\infty$ we have that $\#\overline{G} < +\infty$, then, by applying Artin's lemma, we have that $\overline{K}(E)^{\overline{G}} \subset \overline{K}(E)$ is a finite Galois extension with Galois group G. Now (using lemma 7.1) , one can prove that, since the transcendence degree of $\overline{K}(E)/\overline{K}(E)^{\overline{G}}$ is 1 (by Artin's lemma we have just proved that the extension is finite and algebraic), there exists a unique (up to isomorphisms) smooth curve $\mathcal{C}/K$ and a finite morphism $\phi : E \to \mathcal{C}$ such that $\phi^*(K(\mathcal{C})) = \overline{K}(E)^{\overline{G}}$. To conclude, one has to show that

1. $\mathcal{C}$ is an elliptic curve, considered with neutral element $\phi(O)$;
2. $\ker\phi = G$, this comes from the fact that

$$\ker\phi \cong \mathrm{Aut}(K(E)/\phi^*(K(\mathcal{C}))) = \mathrm{Aut}(K(E)/\overline{K}(E)^{\overline{G}}) = \overline{G} \cong G$$

3. $\phi : E \to \mathcal{C}$ is an isogeny, if we consider over $\mathcal{C}$ the group structure having $\phi(O)$ as neutral element.

[8] dimostra isomorfismo

To prove the first statement, we have to prove that the map ϕ is unramified, i.e. that $\forall Q \in \phi(E)$ then $\#\phi^{-1}(Q) = \deg \phi$. For the properties of finite morphisms of curves, it is always true that $\#\phi^{-1}(Q) \leq \deg \phi$, so let us prove $\geq$. To prove this, we will prove that, if $T \in G$, then $\phi(P+T) = \phi(P)$ for all $P \in E(\overline{K})$. Indeed, for every $f \in K(\mathcal{C})$ one has that

$$f(\phi(P+T)) = (\tau_T^* \circ \phi^*)(f)(P) = \phi^*(f)(P) = f(\phi(P)),$$

where we used the property that τ_T^* fixes every element of $\phi^*(K(\mathcal{C}))$ (since $\phi^*(\overline{K}(\mathcal{C})) = \overline{K}(E)^{\overline{G}}$). Now we can choose as f the coordinate's map, i.e. ϕ_x, ϕ_y and ϕ_z (and we have $\phi_x(T+P) = \phi(P)$ and etc.); this implies that $\phi(P+T) = \phi(P)$ for all P. Hence, if we choose a point $P \in E$ such that $P \in \phi^{-1}(Q)$, then $\phi^{-1}(Q) \supseteq \{P + T \mid T \in G\}$. The points in this set are all distinct, hence

$$\#\phi^{-1}(Q) \geq \#G = \#\overline{G} = [\overline{K}(E) : \overline{K}(E)^{\overline{G}}] = [\overline{K}(E) : \phi^*(K(\mathcal{C}))] = \deg \phi$$

Knowing now that ϕ is unramified, we can apply lemma 7.4 with $\mathcal{C}_1 = E$ and $\mathcal{C}_2 = \mathcal{C}$. Since E is elliptic we have that $g(E) = 1$ and then

$$0 = \deg\phi(2(g(\mathcal{C}) - 1)) \implies g(\mathcal{C}) = 1$$

i.e. E is an elliptic curve. The points 2 and 3 are then clear.

Notice that, if we have another separable isogeny $\phi' : E \to E'$ with the same kernel G, then we can view $\overline{K}(E')$ as a subfield of $\overline{K}(E)$ via the induced embedding $\phi'^* : \overline{K}(E') \to \overline{K}(E)$[9]. Moreover $\phi^*(\overline{K}(E')) \subset \overline{K}(E)^G$; indeed, for every $T \in G$ and every $g \in \overline{K}(E')$,

$$(\tau_T)^*(\phi'^*(g)) = (\tau_T)^*((g \circ \phi')) = (g \circ \phi' \circ \tau_T) \quad \text{and} \quad \phi' \circ \tau_T = \phi' \text{ if } T \in \ker(\phi') = G.$$

Since $\phi' : E \to E'$ is separable, we have $\deg(\phi') = [\overline{K}(E) : \phi'^*(\overline{K}(E'))] = \#G$, so $\phi^*(K(E'))$ must be isomorphic to $\overline{K}(E)^G$ (since they are contained one into the other). Consequently there exists an isomorphism $\iota : \mathcal{C} \to E'$ such that $\phi' = \iota \circ \phi$, hence the isogeny ϕ and the curve $\mathcal{C}$ are unique up to isomorphism. For a detailed proof of this result, see [**?**, Proposition 4.12]. $\square$

[9] Since ϕ is surjective, it is dominant, hence ϕ^* is injecrtive.

8 Teorema di struttura della n-torsione (Teorema 2.7.1)

Lemma 8.1. The $[n]$ endomorphism for an elliptic curve E defined on K with $\mathrm{char}(K) = p$ is separable if and only if $p \not| n$.

Theorem 8.2. Let E be an elliptic curve over a field K and let n be a positive integer. If the characteristic of K does not divide n or it is 0, then

$$E[n] \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

If the characteristic of K is $p > 0$ and $p \mid n$, write $n = p^r n'$ with $p \nmid n'$. Then,

$$E[n] \cong \mathbb{Z}/n'\mathbb{Z} \times \mathbb{Z}/n'\mathbb{Z} \quad \text{or} \quad \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n'\mathbb{Z}.$$

Proof. Assume now that n is not a multiple of the characteristic p of the field; then, from lemma 8.1, we have that the multiplication-by-n map is separable, hence $E[n]$ has order the degree of the endomorphism, which is n^2. Now, applying the structure theorem of finite abelian groups we have that

$$\mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

for some integers $n_1, \ldots, n_k$ with $n_i \mid n_{i+1}$ for all i. Let ℓ be a prime dividing n_1; then $\ell \mid n_i$ for all i; this means that $E[\ell] \subseteq E[n]$ has order ℓ^k; but we have just proved that the order of $E[\ell]$ is ℓ^2, hence $k = 2$. Multiplication by n annihilates $E[n] \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}$, so we must have $n_2 \mid n$. Using that $n^2 = \#E[n] = n_1 n_2$ and that $n_1 | n_2$, therefore, if $p \nmid n$,

$$E[n] \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

It remains to consider the case $p \mid n$. We first determine the p-power torsion on E. By Corollary 2.7.8, the multiplication-by-p is not separable, hence $E[p]$ has order strictly less than the degree of the endomorphism, which is p^2. Since every element of $E[p]$ has order either 1 or p, then the order of $E[p]$ is a power of p, hence it must be 1 or p. If $E[p]$ is trivial, then $E[p^k]$ must be trivial for all k[10]. Assume now that $\#E[p] = p$; then we show that $E[p^k] \cong \mathbb{Z}/p^k\mathbb{Z}$. Notice that under this hypothesis we have that $\deg_{\mathrm{sep}[p]} = p$[11]; now, the multiplication by p^k map is nothing else than the composition of the multiplication by p k times, and we showed that the separability degree is multiplicative with respect to the composition of isogenies; hence this implies that

$$\deg_{\mathrm{sep}}[p^k] = (\deg_{\mathrm{sep}}[p])^k = p^k,$$

which implies that $\#E[p^k] =^{(\text{theorem}6.2)} \deg_{\mathrm{sep}}[p^k] = p^k$ for all $k \geq 1$. Let us now show that there exists an element of order exactly p^k. Assume that there exists an element P of order p^j; since the multiplication-by-p is surjective, there exists a point Q with $pQ = P$. Since $p^j Q = p^{j-1}P \neq \mathcal{O}$ and $p^{j+1}Q = p^j P = \mathcal{O}$, we have that Q has order exactly p^{j+1}. By

[10] If exists $\alpha \in E[p^k]$ then $\alpha^k = 0$ then $< \alpha > \cong \mathbb{Z}/p^k\mathbb{Z}$, and since the last one is cyclic, we have that exists $\beta \in < \alpha >$ with $\beta^p = 0$. And this is impossible.

[11] If $p|n$ then $[p]$ is non-separable; hence $[p] = \alpha \circ \pi$ and since $\deg\pi = p$ we have that $\deg_{sep}([p]) = \deg\alpha = p$.

induction this implies that there are points of order p^k for every k, and so $E[p^k] \cong \mathbb{Z}/p^k\mathbb{Z}$. We can now deal with the general case. Let us write $n = p^r n'$, with $p \nmid n'$; then

$$E[n] \cong {}^{12}E[p^r] \times E[n'].$$

Now from the previous part we have that $E[n'] \cong \mathbb{Z}/n'\mathbb{Z} \times \mathbb{Z}/n'\mathbb{Z}$, and $E[p^r]$ is either 0 or $\mathbb{Z}/p^r\mathbb{Z}$. Using that $\mathbb{Z}/n'\mathbb{Z} \times \mathbb{Z}/p^r\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z}$, we have

$$E[n] \cong \mathbb{Z}/n'\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \quad \text{or} \quad \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

$\square$

[12] This can be prooved using the theorem on the free product of subgroup of a group.

9 La mappa di moltiplicazione per n è separabile se e solo se char(K)=0 oppure char(K) $\nmid$ n (Corollario 2.7.8)

Lemma 9.1. We have that

$$\phi_n(x) = x^{n^2} + \text{ lower degree terms}$$

$$\psi_n^2(x) = n^2 x^{2n^2-1} + \text{ lower degree terms}$$

Theorem 9.2. *Let E be an elliptic curve over a field K. Then the multiplication by n endomorphism $[n] : E \to E$ is separable if and only if n is not divisible by the characteristic of K.*

Proof. We know that the multiplication by n map is given by the formula

$$nP = \left(\frac{\phi_n(x)}{\psi_n^2(x)}, \frac{\omega_n(x,y)}{\psi_n^3(x)} \right).$$

$\Leftarrow$ If n does not divide the characteristic of K then the leading term of $\phi_n(x)'$, i.e. $n^2 x^{n^2-1}$, is not zero and therefore

$$\left(\frac{\phi_n(x)}{\psi_n^2(x)} \right)' \neq 0,$$

which implies that $[n]$ is a separable endomorphism. $\Rightarrow$ On the other hand if $n \mid \operatorname{char} K$ then the leading coefficient of $\psi_n^2(x)$ vanishes and $\deg \psi_n^2$ is less than $n^2 - 1$. This implies that the kernel of $[n]$, which consists of the point $\mathcal{O}$ and the points (x_0, y_0) for which $\psi_n(x) = 0$ (corollary 5.4), is strictly smaller than its degree, which is n^2. This shows that in this case $[n]$ is therefore inseparable as wanted (theorem 6.2). $\square$

10 Teorema di esistenza e unicità dell'isogenia duale (Teorema 2.9.1

Theorem 10.1. For any isogeny $\alpha : E_1 \to E_2$ there exists a unique isogeny $\hat{\alpha} : E_2 \to E_1$ for which $\hat{\alpha} \circ \alpha = [n]$, where $n = \deg \alpha$.

Proof. Let us first prove uniqueness. If $\alpha_1 \circ \alpha = \alpha_2 \circ \alpha$, then $\alpha_1(P) = \alpha_2(P)$ for all $P \in E_2(\overline{K})$ since α is surjective. This implies that the rational maps of α_1 and α_2 must be the same, since they agree on infinitely many points $P \in E_2(\overline{K})$.
Let us now prove existence. First, notice that, if $n = 1$, then α is an isomorphism, hence if we take $\hat{\alpha} = \alpha^{-1}$. Let us now assume $n \geq 2$; we prove the claim by induction on the number of prime factors of n (counted with multiplicity). Let us denote by p the characteristic of the field K.
First, assume that α has prime degree $\ell \neq p$, then α is separable and $\alpha(E_1[\ell])$ is a subgroup of cardinality

$$\#E_1[\ell]/\deg(\alpha) = \deg([\ell])/\deg(\alpha) = \ell^2/\ell = \ell$$

by applying the first isomorphism theorem to $\alpha|_{E_1[\ell]}$ (recall that $\ker \alpha \subseteq E_1[\ell]$). Let $\alpha' : E_2 \to E_3$ be the separable isogeny with $\ker \alpha' = \alpha(E_1[\ell])$ (i can use the theorem since i proved that the cardinality of the ker is finite); then, $\ker(\alpha' \circ \alpha) = \{P \in E_1 \mid \alpha(P) \in \ker(\alpha')\} = E_1[\ell]$, and since $[\ell] : E_1 \to E_1$ is another isogeny with the same kernel, there exists an isomorphism $\iota : E_3 \to E_1$ such that $\iota \circ \alpha' \circ \alpha = [\ell]$; if we put $\hat{\alpha} := \iota \circ \alpha'$ we have $\hat{\alpha} \circ \alpha = [\ell]$ as wanted.
Let us now assume that α has prime degree equal to the characteristic p; hence there are two cases.
Case 1: If α is separable we have that $\#\ker\alpha = \deg\alpha$, then $\ker \alpha$ is a subgroup of $E_1[p]$ of order p, which is the largest possible size of $E_1[p]$ in characteristic p (from the theorem of strucutre), so $\ker \alpha = E_1[p] \cong \mathbb{Z}/p\mathbb{Z}$ and $\deg_s[p] = p$ (in particular notice that this implies that the curve E_1 in this case is ordinary). Now $\deg[p] = p^2$, hence by lemma 6.1 we have that $[p] = \alpha' \circ \pi$ with $\pi : E_1 \to E_1^{(p)}$ the p-power Frobenius morphism and $\alpha' : E_1^{(p)} \to E_1$ a separable isogeny. If we call $\pi_2 : E_2 \to E_2^{(p)}$, then we have that $\pi_2 \circ \alpha = \alpha^{(p)} \circ \pi_1$ where $\alpha^{(p)} : E_1^{(p)} \to E_2^{(p)}$ is obtained by replacing each coefficient of α by its pth power; indeed, if α is given by

$$\alpha(x, y) = \left(\frac{u(x)}{v(x)}, \frac{s(x)}{t(x)} y\right),$$

then

$$\pi_2(\alpha(x, y)) = \left(\frac{u(x)^p}{v(x)^p}, \left(\frac{s(x)}{t(x)}\right)^p y^p\right) = \left(\frac{u^{(p)}(x^p)}{v^{(p)}(x^p)}, \left(\frac{s^{(p)}(x^p)}{t^{(p)}(x^p)}\right) y^p\right) = \alpha^{(p)}(\pi_1(x, y)).$$

We have then

$$\ker(\alpha^{(p)} \circ \pi_1) = \ker(\pi_2 \circ \alpha) = \ker \alpha = \ker[p] = \ker(\alpha' \circ \pi_1),$$

since the Frobenius morphisms π_1 and π_2 have trivial kernel, and it follows that $\alpha^{(p)}$ and α' are separable isogenies [13] with the same kernel; there is thus an isomorphism $\iota : E_2^{(p)} \to E_1$

[13] $\alpha^{(p)}$ is separable since $\left(\frac{u^{(p)}(x)}{v^{(p)}}\right)' \neq 0$.

such that $\alpha' = \iota \circ \alpha^{(p)}$. If we now put $\hat{\alpha} = \iota \circ \pi_2$ then

$$\hat{\alpha} \circ \alpha = \iota \circ \pi_2 \circ \alpha = \iota \circ \alpha^{(p)} \circ \pi_1 = \alpha' \circ \pi_1 = [p],$$

as wanted.

Case 2: If α is inseparable then we must have $\alpha = \iota \circ \pi$ for some isomorphism ι. From the strucutre theorem we have two cases, if $E[p] = \{\mathcal{O}\}$, then $[p]$ is purely inseparable of degree p^2, so $[p] = \iota' \circ \pi^2$ for some isomorphism ι', and we may take $\hat{\alpha} = \iota' \circ \pi \circ \iota^{-1}$. If $E[p] \cong \mathbb{Z}/p\mathbb{Z}$, then $[p] = \alpha' \circ \pi$ for some separable isogeny α' of degree p and we may take $\hat{\alpha} = \alpha' \circ \iota^{-1}$.

Let us now treat the case n composite; if so, thanks to Exercise 2.6.5, we can decompose α into a sequence of isogenies of prime degree; in particular we can write $\alpha = \alpha_1 \circ \alpha_2$, where $\deg \alpha_i = n_i < n$ and $n_1 n_2 = n$. Let now $\hat{\alpha} := \hat{\alpha}_2 \circ \hat{\alpha}_1$, where the existence of the $\hat{\alpha}_i$ is given by the inductive hypothesis. Then

$$\hat{\alpha} \circ \alpha = (\hat{\alpha}_2 \circ \hat{\alpha}_1) \circ \alpha = \hat{\alpha}_2 \circ (\hat{\alpha}_1 \circ \alpha_1) \circ \alpha_2 = \hat{\alpha}_2 \circ [n_1] \circ \alpha_2 = [n_1] \circ \hat{\alpha}_2 \circ \alpha_2 = [n_1][n_2] = [n],$$

where we used the property that $[n_1] \circ \alpha_2 = \alpha_2 \circ [n_1]$. This concludes the proof. $\square$

11 Teorema di Hasse (Teorema 3.1.2)

Lemma 11.1. Let $\pi_E : E \to E$ the Frobenius endomorphism; then, the map $\pi_E - [1]$ is separable.

Definition 11.2. Let A an abelian group; then, $d : A \to R$ is said to be a quadratc form if it satisfies the following properties:

1. $d(a) = d(-a)$ for all $a \in A$;
2. the function $A \times A \to \mathbb{R}$ such that $(a, b) \mapsto d(a, b) - d(a) - d(b)$ is bilinear.

Moreover, we say that is positive definite if

- $d(a) = 0$ if and only if $a = 0$;
- $d(a) \geq 0$ for all $a \in A$

Lemma 11.3. The map deg: $\mathrm{Hom}(E_1, E_2) \to \mathbb{Z}$ is a positive definite quadratic form.

Lemma 11.4. Let $d : A \to \mathbb{Z}$ be a positive definite quadratic form; then, for every $\phi, \psi \in A$

$$|d(\phi + \psi) - d(\phi) - d(\psi)| \leq 2\sqrt{d(\phi)d(\psi)}$$

Theorem 11.5. Let $E/\mathbb{F}_q$ be an elliptic curve over a finite field; then,

$$E(\mathbb{F}_q) = q + 1 - t$$

where t is the trace of the Frobenius endomorphism π_E and $|t| \leq 2\sqrt{q}$.

Proof. Recall that $\mathbb{F}_q$ is defined as the splitting field of the polynomial $x^q - x$ over $\mathbb{F}_q$, where$p = char(\mathbb{F}_q)$; thus $\mathbb{F}_q$ is precisely the subfield of the q-power Frobenius automorphism $x \mapsto x^q$, therefore

$$E(\mathbb{F}_q) = \{P \in E(\overline{\mathbb{F}_q}) : \pi_E(P) = P\} = ker(\pi_E - 1)$$

where 1 denotes the multiplication-by-1 map $[1] \in End(E)$ (i.e. the identity). The morphism $\pi_E - 1$ is separable by lemma 11.4, thus the cardinality of its kernel is equal to its degree. But now

$$[deg(\pi_E - 1)] = (\widehat{\pi_E - 1}) \circ (\pi_E - 1) = (\widehat{\pi_E} - 1) \circ (\pi_E - 1) = \widehat{\pi_E} \circ \pi_E + 1 - (\widehat{\pi_E} + \pi_E) = [q + 1 - t].$$

It remains only to show that $|t| \leq 2\sqrt{q}$. To prove this, we use the fact that the degree map on $\mathrm{End}(E)$ is a positive definite quadratic form, hence by lemma 11.4

$$|t| = |\#E(\mathbb{F}_q) - q - 1| = |deg(\pi E - 1) - deg(\pi_E) - deg([1])| \leq 2\sqrt{deg(\pi_E)deg([1])} = 2\sqrt{q}$$

which concludes the proof. $\square$